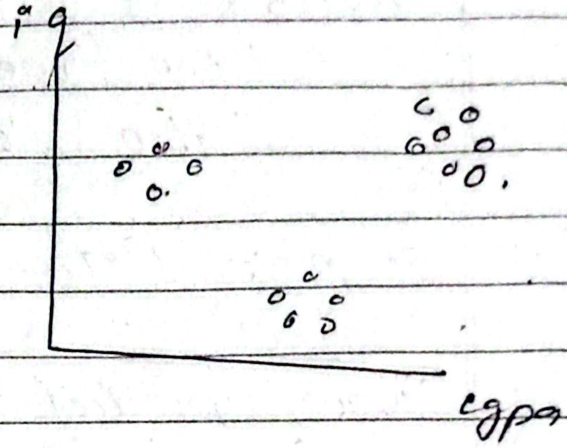
K-Means Clustering:

Steps:

1) Decide no. of clusters.
K = 3.



2) Initialize centroids.
[center of clusters]

3) Assign cluster [by calculating eculidian distance].

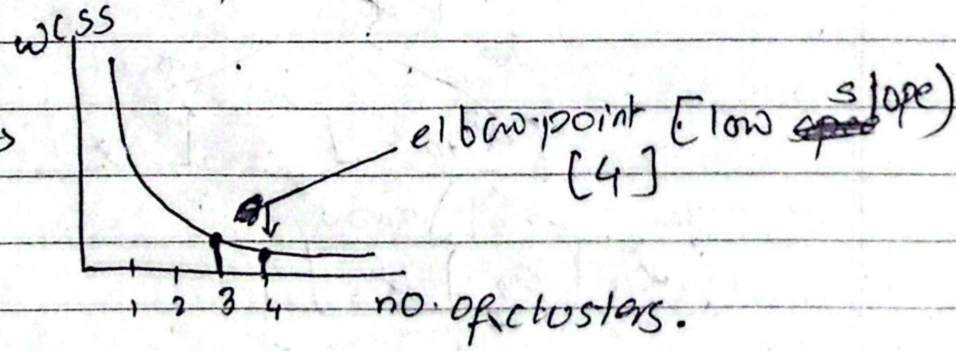
[reassign occur of data in new cluster]

4) Calculate centroid again.

5) Finish.

Elbow Method: [to find optimum value of k]

elbow curve →



$d_1^2 + d_2^2 + ---$

$wcss_1 > wcss_2 > wcss_3 > --- - wcss_n$.

max. no. of clusters = total no. of points [rows]